

Chapter 1

Understanding Cloud Computing

1.1 Introduction to Cloud Computing

1.1.1 Introduction to cloud computing

1.1.2 The power of cloud

- Vertical and Horizontal Scaling
- Pay as you go pricing
- Speed
- Reliability
- Security

1.1.3 Cloud service models

- IaaS: owning car
- PaaS: ridesharing
- SaaS: taking a bus
- FaaS: Function as a Service, auth and transaction
- XaaS: Anything as a Service, covers everything

1.2 Cloud Deployment

1.2.1 Cloud Deployment models

- Private cloud: uses virtualization for on demand resources and can be located off-premises, direct control over resources available, and how data is stored
- Public cloud: internet accessible for general public, no access to data center and hardware
- Hybrid cloud: mix, for seasonal spikes or cloud bursting
- Multicloud : combine different CSP
- Community cloud: shared by specific community with common interest

1.2.2 Regulations on the cloud

1.2.3 Cloud computing roles

1.3 Cloud Providers and Case Studies

1.3.1 An overview of providers

1.3.2 Amazon Web Services

1.3.3 Microsoft Azure

1.3.4 Google Cloud